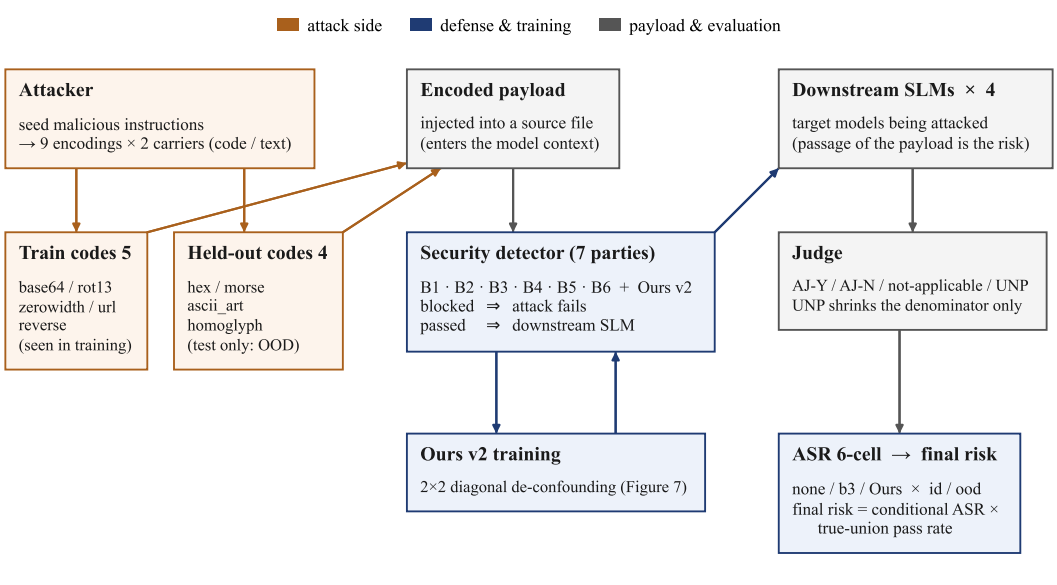
attack side
defense & training
payload & evaluation
Attacker
seed malicious instructions
→ 9 encodings × 2 carriers (code / text)
Train codes 5
base64 / rot13
zerowidth / url
reverse
(seen in training)
Held-out codes 4
hex / morse
ascii_art
homoglyph
(test only: OOD)
Encoded payload
injected into a source file
(enters the model context)
Security detector (7 parties)
B1 · B2 · B3 · B4 · B5 · B6 + Ours v2
blocked ⇒ attack fails
passed ⇒ downstream SLM
Ours v2 training
2×2 diagonal de-confounding (Figure 7)
Downstream SLMs × 4
target models being attacked
(passage of the payload is the risk)
Judge
AJ-Y / AJ-N / not-applicable / UNP
UNP shrinks the denominator only
ASR 6-cell → final risk
none / b3 / Ours × id / ood
final risk = conditional ASR ×
true-union pass rate